
INTERNATIONAL GCSE BIOLOGY 9201/2 Paper 2

Mark Scheme

November 2018

Version: 1.0 Final



1 8 B Y 9 2 0 1 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
01.1	(sun)light	allow sun	1	AO1 3.3.1 a	E
01.2	population of grass would increase	allow grow more/taller	1	AO2 3.3	E
	population of snakes would decrease	allow snakes find another food source	1		
01.3	880 (kg)		1	AO2 6.3 (3) 3.3.1 b	G
01.4	3-layered triangular pyramid		1	AO2 3.3.1 c	E
	all three labels in food chain order where grass is widest layer	allow correct order but upside down	1		
01.5	mice producing faeces (droppings)		1	AO2 3.3.1 b	A
	mice respiring		1		
Total			8		

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
02.1	parasite		1	AO1 3.3.2 d	G
02.2	Feature Body sections produce many eggs Hooks and suckers Long and flattened shape Thick cuticle How the feature helps the tapeworm survive increases the rate of the food absorption from the host protects tapeworm from damage by host's digestive enzymes increases the chance of infecting a new host provides a firm grip on the intestine wall	3 marks for 4 correct 2 marks for 3 correct 1 mark for 1 or 2 correct	3	AO2 3.3.2 d	G
02.3	(tapeworms) absorb digested food from (intestine of) person/host (so) less food available for person/host (to absorb) (and) energy intake is reduced or energy stores are used (so human loses weight)	allow any sensible symptom that would cause reduced absorption, eg gut irritation, nausea, diarrhoea, loss of appetite for 1 mark	1 1 1	AO2 AO3 3.4.4a 3.3.2d	E
02.4	harm/damage/kill humans	allow any correct example of harm	1	AO3 3.3.2d	E
02.5	number of (MRSA) bacteria		1	AO2 6.1	E

02.6	Drug A number of bacteria increased (rate of) increase reduced after 2 hours Drug B number of bacteria stayed the same for the first hour then (number of bacteria) decreased (after 1 hour)	allow appropriate use of figures to illustrate the trends	1 1 1 1	AO3 3.4.7l,j 6.1	E
02.7	for a valid conclusion or to show results/difference is only due to the drugs being tested		1	AO4 6.1	E
02.8	(grow MRSA bacteria in nutrient broth) with water/no drug added		1	AO4 6.1	E
Total			15		

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
03.1	microscope	ignore hand lens/magnifying glass ignore any qualification of microscope	1	AO3 6.1	G
03.2	$(0.2 \times 0.2) = 0.04 \text{ (mm}^2\text{)}$ $(8 \div 0.04) = 200$	200 (per mm ²) scores 2 marks allow 8 ÷ by answer from mp1 for 1 mark	1 1	AO2 6.3 (14,3)	E
03.3	guard cells		1	AO1 3.2.2 e	A
03.4	any two from: - limits/controls water loss - prevents plant wilting - controls/allows entry of CO₂ (for photosynthesis) - controls/allows exit of O₂ (from photosynthesis)	allow controls gas exchange for 1 mark	2	AO1 3.2.2d 3.2.2c	E
03.5	Method 1 $\frac{2 \times 74 \times 49}{3}$ 2417.33 2417	2417 (mm ²) scores 3 marks	1 1 1	AO2 6.3 (4, 14)	E

03.6	any two from: • less accurate • shape of leaf may not be oval • edges of leaves may be indented/irregular • depends on where measurements are taken from		2	AO4 6.1	E
03.7	prevent water loss/evaporation from surface of flask/water		1	AO4 6.1	E
03.8	1988 × (14 + 122) 270368 2.7 × 10 ⁵	2.7 × 10 ⁵ scores 3 marks 2.7(0368) × 10 ⁵	1 1 1	AO2 6.3 (15)	E
03.9	Level 3: The plan would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	5-6	AO3 3.2.2 d 6.1	E	
	Level 2: The plan would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3-4	AO2 AO3 3.2.2 d 6.1	E	
	Level 1: The plan would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1-2	AO2 3.2.2 d 6.1	E	
	No relevant content	0			
	Indicative content • use the same leaf or leaves with same SA or leaves from the same plant • same light intensity • same temperature/humidity • keep distance of fan the same or keep fan speed the same • change fan speed a minimum 3 x or change distance of fan a minimum 3 x • measure mass of flask/apparatus before and after test • leave for same length of time • calculate the difference in mass • repeat and calculate a mean for each fan speed or fan distance • produce graph or chart to show relationship				
Total			21		

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
04.1	lipase		1	AO1 3.2.4 c	A
04.2	to ensure all chemicals/reagents are at the correct/same/at 30 °C temperature (before mixing)		1	AO4 6.1	E
04.3	any two from: - allows elimination/identification of anomalies - allows a more accurate mean to be calculated - can compare repeated results/check patterns are the same/allows for repeatability	do not allow reproducibility	2	AO4 6.1	E
04.4	use pH meter/probe		1	AO4 6.1	E
04.5	any two from: - record time in smaller intervals - use a thermostatically controlled water bath - use the same volume of indicator - include a control of water with milk and indicator - use same volume of enzyme/bile/liquids (in tubes) - use same concentration of enzyme/bile - repeat investigation at 37 °C - stir/mix contents of tubes	allow use more precise measurements for one mark	2	AO4 6.1	E

04.6	(Prediction:) no change (after 20 minutes) or more than 20 minutes		1	AO3 6.1 AO2 3.2.4 b	E
	(Explanation:) enzyme denatured or substrate/fat no longer fits active site or shape of active site changed	do not allow enzyme killed	1		
	(so) pH of mixture does not change or no fatty acids produced	allow active site damaged/destroyed	1		

04.7	bile reduces the time taken for the enzyme to break down fats or bile increases the enzyme's rate of reaction		1	AO3 6.1 3.2.4 c	E
	bile does not (chemically) digest fats		1		
	boiling bile has no effect upon its activity	allow bile is not a protein/enzyme	1		

Total			13
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Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
05.1	$\frac{0.9 + 1.7 + 1.2 + 2.8 + 2.4}{5}$ 1.8 (cm)	1.8 (cm) scores 2 marks	 1 1	 AO2 6.1 6.3	 E
05.2	Level 3: Relevant points (reasons/causes) are identified, given in detail and logically linked to form a clear account.		5-6	AO3 3.2.4 c 3.4.5 a,b,c	E
	Level 2: Relevant points (reasons/causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.		3-4	AO3 3.2.4 c 3.4.5 a,b,c	E
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logically linking.		1-2	AO2 3.2.4 c 3.4.5 a,b,c	E
	No relevant content		0		
	Indicative content • (damaged pancreas so) less/no insulin/glucagon produced • fewer pancreatic cells to detect levels of blood glucose • so cannot monitor/control blood glucose levels or have diabetes • (bile duct blocked so) bile cannot reach small intestine • so reduced fat digestion or so fats not emulsified • pH of small intestine acidic • so lipase does not work efficiently • (bile duct blocked so digestive) enzymes cannot reach small intestine • (damaged pancreas so) no/less digestive enzymes produced • so reduced carbohydrate/fat/protein digestion for full marks must include effect upon digestion and blood glucose.				

Total			8
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Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
06.1	joining/fusion of male and female gametes (so) half of the genes/chromosomes come from each parent	allow each gamete has a unique set of genes from parent	1 1	AO1 AO2 3.5.1 a	E
06.2	any three from: • splitting apart of cells (from embryo) • before they become specialised • each (new) cell divides (by mitosis) to form an embryo • embryos placed into host mother's womb	allow correct description of an 8/16 cell stage	3	AO1 3.5.5 a	E
06.3	produce several/many embryos		1	AO3 3.5.5 a	E
06.4	no/little genetic variation (and) less chance that useful characteristic/genes/alleles will be present (so) less likely to survive changes in environment or natural selection cannot occur	allow described changes (eg new predator or new disease or different food source) allow if raised in captivity unable to survive in wild for 1 mark	1 1 1	AO1 AO2 3.6.2 b	E
06.5	undifferentiated/ non-specialised cells		1	AO1 3.5.2 k	E

06.6	any five from: advantages: • useful when rhinos are unable to breed naturally • prevents extinction or continues genetic line • use of both (IVF and stored tissue) would improve the genetic diversity disadvantages: • not cost effective • not proven to work • limited gene pool/only using genes from limited individuals/not genetically diverse • offspring unlikely to be able to survive in the wild • diverts attention/money from habitat loss and stopping poaching • may harm/kill remaining rhinoceroses	For full marks must include at least one advantage and one disadvantage	5	AO3 3.5.2 k 3.5.1	E
Total			15		

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
07.1	skin thermoregulatory centre/hypothalamus	ignore brain	1 1	AO1 3.4.4 a,b	E
07.2	(plastic) prevents evaporation (and) evaporation cools body (so this) reduces/stops transfer of (thermal) energy (from skin) to environment	allow reduces heat loss or body temperature less likely to fall below 35 °C	1 1 1	AO1 AO2 3.4.4 c	E
07.3	any five from: • less shivering and muscle <u>contraction</u> • (so) less respiration and released thermal energy • blood vessels supplying the skin capillaries do not constrict (effectively) • blood flow to capillaries in the skin is not reduced • greater energy transfer to the surroundings • (so) baby cannot maintain temperature due to the cumulative effect of greater heat loss and reduction in ability to release heat (from respiration)		5	AO2 3.4.4 d	E
Total			10		